

BANDIT

➤ **Level 0**

Level Goal

The goal of this level is for you to log into the game using SSH. The host to which you need to connect is `bandit.labs.overthewire.org`, on port 2220. The username is `bandit0` and the password is `bandit0`. Once logged in, go to the Level 1 page to find out how to beat Level 1.

Command Used : ssh

- ✓ In terminal i run the command **ssh bandit0@bandit.labs.overthewire.org -p 2220** as it is mentioned in level goal

[illegible]

- ✓ Putting password “bandit0” as it is mentioned in the level goal and we are in !

```
--[ Tips ]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32                compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro       disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

bandit0@bandit:~$
```

➤ Level 0 → Level 1

Level Goal

The password for the next level is stored in a file called `readme` located in the home directory. Use this password to log into `bandit1` using SSH. Whenever you find a password for a level, use SSH (on port 2220) to log into that level and continue the game.

Command Used : `ls`, `cat`

- ✓ I used “`ls`” command in terminal to list the files and folders.
- ✓ After running “`ls`” command i got a `readme` file.
- ✓ I used “`cat`” command to see in the file. So `readme` file contain next level password

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: 6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR
bandit0@bandit:~$
```

Password: 6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR

➤ Level 1 → Level 2

Level Goal

The password for the next level is stored in a file called `-` located in the home directory

Command Used : `ssh`, `ls`, `cat ./-`

- ✓ As we have password for bandit1 so we run the command “**`ssh bandit1@bandit.labs.overthewire.org -p 2220`**”
- ✓ after that put the password : `6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR`

```
husnain@husnain ~$ ssh bandit1@bandit.labs.overthewire.org -p 2220
bandit1@bandit.labs.overthewire.org:~$
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit1@bandit.labs.overthewire.org's password:
```

- ✓ Used “`ls`” command to view files and folders
- ✓ got a “`-`” file in this level
- ✓ i used google to learn how to open a “`-`” file using “`cat`” command and the command is “`cat ./-`”

```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ cat ./-
PK8fYLZg2hnHSz83pIBL1iEPKdD3QToB
bandit1@bandit:~$
```

Password: PK8fYLZg2hnHSz83pIBL1iEPKdD3QToB

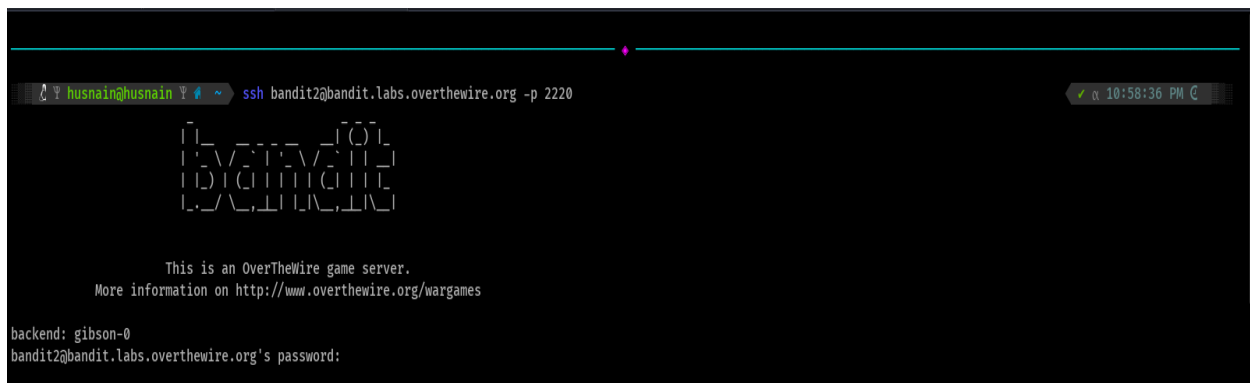
➤ Level 2 → Level 3

Level Goal

The password for the next level is stored in a file called --spaces in this filename-- located in the home directory

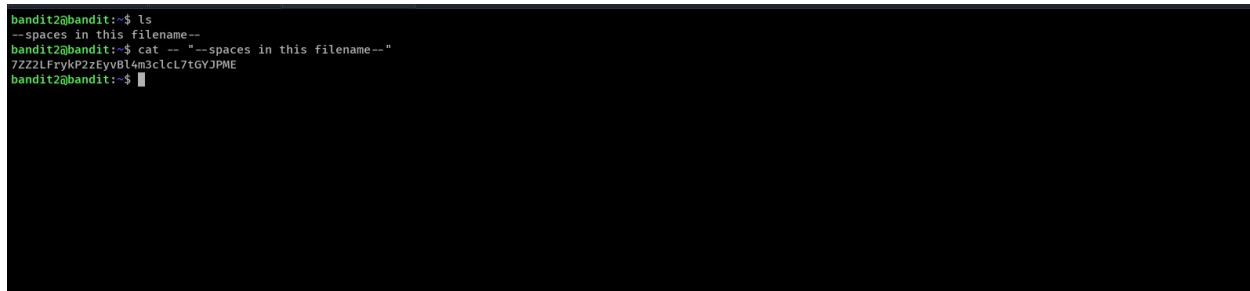
Command Used : ssh, ls, cat with flag “- -” to view the file

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command “**ssh bandit2@bandit.labs.overthewire.org -p 2220**” to logged in bandit2
- ✓ Used password : PK8fYLZg2hnHSz83pIBL1iEPKdD3QToB



```
husnain@husnain ~$ ssh bandit2@bandit.labs.overthewire.org -p 2220
bandit2@bandit.labs.overthewire.org:~$
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit2@bandit.labs.overthewire.org's password:
```

- ✓ List the files and folders.
- ✓ Got a file - - spaces in the filename - -.
- ✓ Used google to learn how to open a file with spaces in the filename.
- ✓ Used command cat - - “- - spaces in the filename - -” to open it and got the password for next level



```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat -- "--spaces in this filename--"
7ZZ2LFrykP2zEyvBl4m3clcl7tGYJPME
bandit2@bandit:~$
```

Password: 7ZZ2LFrykP2zEyvBl4m3clcl7tGYJPME

➤ **Level 3 → Level 4**

Level Goal

The password for the next level is stored in a hidden file in the inhere directory.

Command Used : ssh, ls, cd,

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command “**ssh bandit3@bandit.labs.overthewire.org -p 2220**” to logged in bandit3
- ✓ Used password : 7ZZ2LFrykP2zEyvBl4m3clcl7tGYJPME

```
husnain@husnain ~$ ssh bandit3@bandit.labs.overthewire.org -p 2220
```

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

backend: gibson-0
bandit3@bandit.labs.overthewire.org's password:

- ✓ List the files and folders using “ls”
- ✓ Got a folder “inhere”.
- ✓ Used “cd” command to get in the folder.
- ✓ Used “ls” command to list all directories, found nothing.
- ✓ Then used “ls -la” to see also hidden files.
- ✓ Got a hidden file “...Hiding-From-You”
- ✓ Then used command “cat ../Hiding-From-You” to open it and got the password for next level

```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere/
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -la
total 12
drwxr-xr-x 2 root root 4096 Jun 24 14:59 .
drwxr-xr-x 3 root root 4096 Jun 24 14:59 ..
-rw-r-- 1 bandit4 bandit3 33 Jun 24 14:59 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
xzTXq1rDJQVVAzdv5Chq1TQytWuFAMq
bandit3@bandit:~/inhere$
```

Password: xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq

➤ Level 4 → Level 5

Level Goal

The password for the next level is stored in the only human-readable file in the inheres directory. Tip: if your terminal is messed up, try the “reset” command.

Command Used : ssh, ls, cd, cat

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command “**ssh bandit4@bandit.labs.overthewire.org -p 2220**” to logged in bandit4
- ✓ Used password : xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq

```
hussain@hussain ~$ ssh bandit4@bandit.labs.overthewire.org -p 2220
bandit4@bandit.labs.overthewire.org:~$
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit4@bandit.labs.overthewire.org's password: 
```

- ✓ List the files and folders using “ls”
- ✓ Got a folder “inhere”.
- ✓ Used “cd” command to get in the folder.
- ✓ Used “ls” command to list all directories, found many files.
- ✓ Then used a command to open all file at once by putting “*” at end — “cat ./-file*”, it will open all files from 0 to onward all files
- ✓ After that i saw there is encrypted or dummy data in some files but in fifth line there is password for the next level

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere/
bandit4@bandit:~/inhere$ ls
-file00 -file01 -file02 -file03 -file04 -file05 -file06 -file07 -file08 -file09
bandit4@bandit:~/inhere$ cat ./-file*
Y1[eb10000 00>0D-1K0
3'0000kbj.j000W0000000y1b3009az000090000rf0VE0fQ0|00000000000[]000y0R000f000Y00D00x)00z0TAQ0c0a0u~00[000Z]000dS00 N10000;(1-500Iv0L0;M0b{0#00LSv
++3G
0'000E'4000g0~0~0'D
06C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
F5000000h0L000000
0!0}0LWk0j0s000!Z600000?H"093jXA0,08000z0bandit4@bandit:~/inhere$ 
```

Password: 6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG

➤ **Level 5 → Level 6**

Level Goal

The password for the next level is stored in a file somewhere under the inhere directory and has all of the following properties:

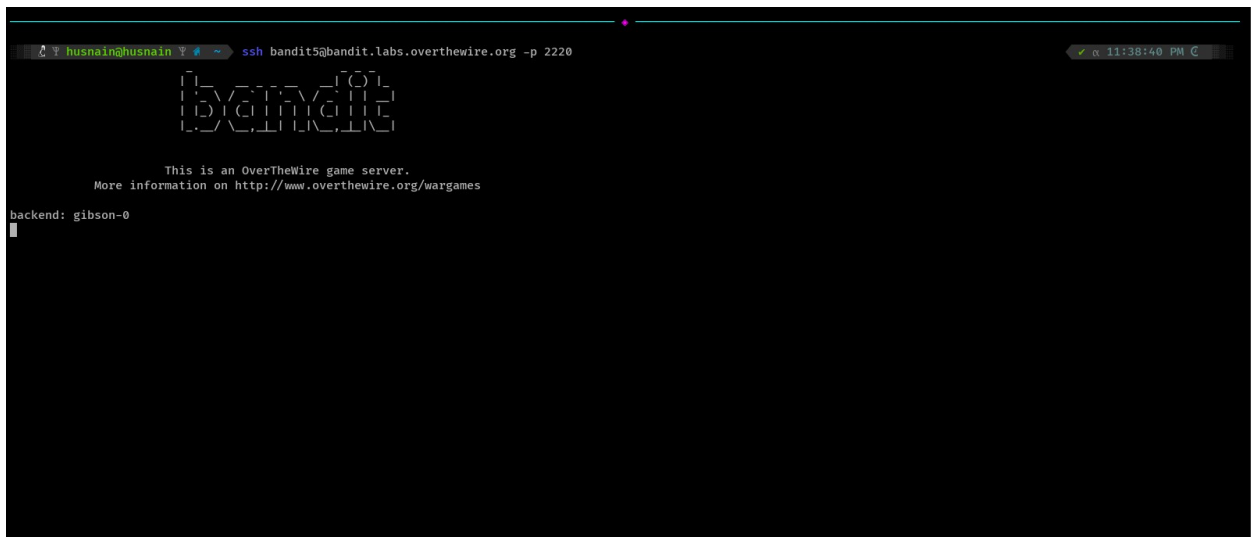
human-readable

1033 bytes in size

not executable

Command Used : ssh, ls, cd, find, cat

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command “**ssh bandit5@bandit.labs.overthewire.org -p 2220**” to logged in bandit5
- ✓ Used password : 6C7h9GD8M6ai5nr7wo1RonrzFjj9ylrG



- ✓ List the files and folders using “ls”
- ✓ Got a folder “inhere”.
- ✓ Used “cd” command to get in the folder.
- ✓ Used “ls” command to list all directories, found many folders.
- ✓ Instead of looking all folders and their files one by one, i used find command “find . -type f -size 1033c” to search the file which size is 1033 bytes as mentioned in level goal. (-type f means “find regular files” and c at end of 1033 means exact 1033 bytes)
- ✓ After running find command i got the file with 1033 bytes in the maybehere07 folder and file name is “.file2”.
- ✓ I go to the maybehere07 folder then run “ls”, so i did not see the “.file2”.

- ✓ As i know when file name is start with dot then it must be hidden so i used “ls -la” command to list all hidden files, after that i saw the .file2.
- ✓ Used cat command to view the file and got the password for the next level.

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere/
bandit5@bandit:~/inhere$ ls
maybehere00 maybehere02 maybehere04 maybehere06 maybehere08 maybehere10 maybehere12 maybehere14 maybehere16 maybehere18
maybehere01 maybehere03 maybehere05 maybehere07 maybehere09 maybehere11 maybehere13 maybehere15 maybehere17 maybehere19
bandit5@bandit:~/inhere$ find . -type f -size 1033c
./maybehere07/.file2
bandit5@bandit:~/inhere$ cd maybehere07/
bandit5@bandit:~/inhere/maybehere07$ ls
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3
bandit5@bandit:~/inhere/maybehere07$ cat file2
cat: file2: No such file or directory
bandit5@bandit:~/inhere/maybehere07$ ls 0la
ls: cannot access '0la': No such file or directory
bandit5@bandit:~/inhere/maybehere07$ ls -la
total 56
-rwxr-x-- 1 root bandit5 3663 Jun 24 14:59 -file1
-rw-r--r- 1 root bandit5 2488 Jun 24 14:59 -file2
-rwxr-x-- 1 root bandit5 3362 Jun 24 14:59 -file3
drwxr-x-- 2 root bandit5 4096 Jun 24 14:59 .
drwxr-x-- 22 root bandit5 4096 Jun 24 14:59 ..
-rwxr-x-- 1 root bandit5 3065 Jun 24 14:59 .file1
-rw-r--r- 1 root bandit5 1033 Jun 24 14:59 .file2
-rwxr-x-- 1 root bandit5 1997 Jun 24 14:59 .file3
-rwxr-x-- 1 root bandit5 4130 Jun 24 14:59 spaces file1
-rw-r--r- 1 root bandit5 9064 Jun 24 14:59 spaces file2
-rwxr-x-- 1 root bandit5 1022 Jun 24 14:59 spaces file3
bandit5@bandit:~/inhere/maybehere07$ cat .file2
pXa26xhMWaC2SvDotA4r9EgZkulOeSBW
bandit5@bandit:~/inhere/maybehere07$
```

Password: pXa26xhMWaC2SvDotA4r9EgZkulOeSBW

➤ Level 6 → Level 7

Level Goal

The password for the next level is stored somewhere on the server and has all of the following properties:

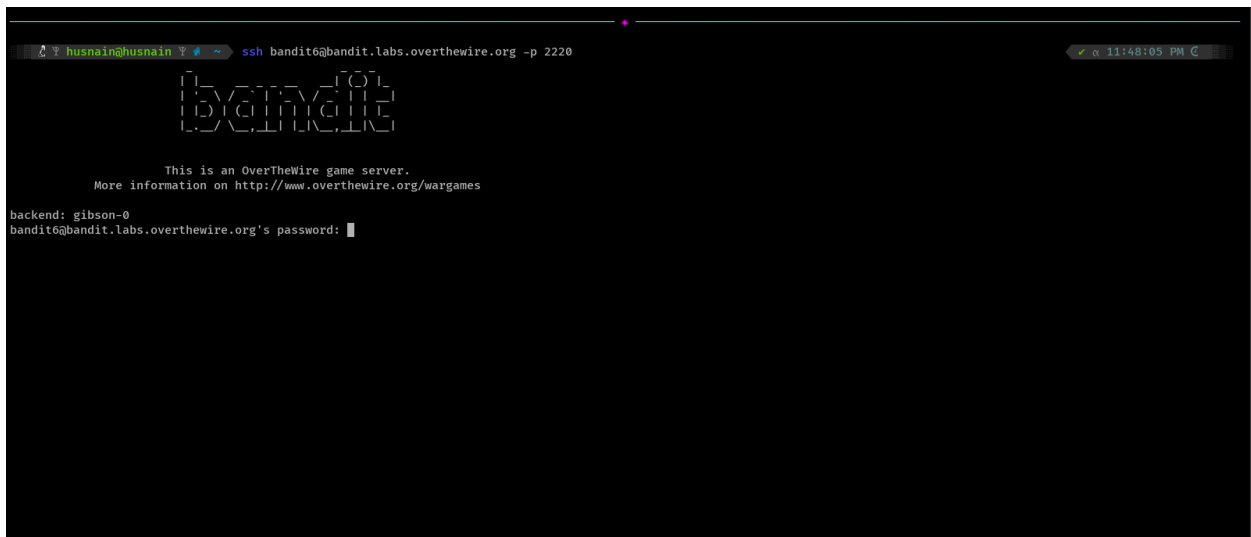
owned by user bandit7

owned by group bandit6

33 bytes in size

Command Used : ssh, ls, ls -la, find, cat

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command "**ssh bandit6@bandit.labs.overthewire.org -p 2220**" to logged in bandit6
- ✓ Used password : pXa26xhMWaC2SvDotA4r9EgZkulOeSBW



```
husnain@husnain ~$ ssh bandit6@bandit.labs.overthewire.org -p 2220
bandit6@bandit.labs.overthewire.org:~$
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
bandit6@bandit.labs.overthewire.org's password: pXa26xhMWaC2SvDotA4r9EgZkulOeSBW
```

- ✓ List the files and folders using "ls"
- ✓ Got nothing then used "ls -la" command but found nothing interested.
- ✓ Review the level goal statement , so their is mentioned that password for next level is on server, owned by user bandit7 , owned by group bandit6 and file size is 33 bytes
- ✓ I asked deepseek to help and it tell me exact command to find the file "**find / -user bandit7 -group bandit6 -size 33c** ", this command give me some files with their permission as output.
- ✓ As there was many files so i used "**2>/dev/null**" flag to skip the error files and this tell me that exact location of the file where the password stored .
- ✓ Used "cat" command to view the file and got the password for the next level.

```
bandit6@bandit:~$ ls
bandit6@bandit:~$ ls -la
total 20
drwxr-xr-x  2 root root 4096 Jun 24 14:58 .
drwxr-xr-x 150 root root 4096 Jun 24 15:02 ..
-rw-r--r--  1 root root  220 Feb 13 12:16 .bash_logout
-rw-r--r--  1 root root 3851 Jun 24 14:50 .bashrc
-rw-r--r--  1 root root  807 Feb 13 12:16 .profile
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c
find: '/snap': Permission denied
find: '/lost-found': Permission denied
find: '/manpage/manpage3-pw': Permission denied
find: '/drifter/drifter14_src/axTLS': Permission denied
find: '/run/pam_pidns': Permission denied
find: '/run/udisks2': Permission denied
find: '/run/chrony': Permission denied
find: '/run/user/11019': Permission denied
find: '/run/user/16002': Permission denied
find: '/run/user/11529': Permission denied
find: '/run/user/11033': Permission denied
find: '/run/user/8002': Permission denied
find: '/run/user/11022': Permission denied
find: '/run/user/16001': Permission denied
find: '/run/user/11010': Permission denied
find: '/run/user/11009': Permission denied
find: '/run/user/11023': Permission denied
find: '/run/user/8003': Permission denied
find: '/run/user/11025': Permission denied
find: '/run/user/11017': Permission denied
find: '/run/user/11032': Permission denied
find: '/run/user/11020': Permission denied
find: '/run/user/11001': Permission denied
find: '/run/user/11004': Permission denied
find: '/run/user/12002': Permission denied
find: '/run/user/11007': Permission denied
find: '/run/user/11003': Permission denied
find: '/run/user/11016': Permission denied
```

```
find: '/proc/2/ns': Permission denied
find: '/proc/66/task/66/fd/6': No such file or directory
find: '/proc/66/task/66/fdinfo/6': No such file or directory
find: '/proc/66/fd/5': No such file or directory
find: '/proc/66/fdinfo/5': No such file or directory
find: '/home/ubuntu': Permission denied
find: '/home/bandit5/inhere': Permission denied
find: '/home/drifter8/chroot': Permission denied
find: '/home/bandit28-git': Permission denied
find: '/home/drifter6/data': Permission denied
find: '/home/bandit29-git': Permission denied
find: '/home/bandit30-git': Permission denied
find: '/home/leviathan0/.backup': Permission denied
find: '/home/bandit31-git': Permission denied
find: '/home/leviathan4/.trash': Permission denied
find: '/home/bandit27-git': Permission denied
find: '/dev/mqueue': Permission denied
find: '/dev/shm': Permission denied
find: '/root': Permission denied
find: '/sys/kernel/tracing/osnoise': Permission denied
find: '/sys/kernel/tracing/hwlat_detector': Permission denied
find: '/sys/kernel/tracing/instances': Permission denied
find: '/sys/kernel/tracing/trace_stat': Permission denied
find: '/sys/kernel/tracing/per_cpu': Permission denied
find: '/sys/kernel/tracing/options': Permission denied
find: '/sys/kernel/tracing/rv': Permission denied
find: '/sys/kernel/debug': Permission denied
find: '/sys/fs/pstore': Permission denied
find: '/sys/fs/bpf': Permission denied
find: '/tmp': Permission denied
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /v
var/ vortex/
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
Bmnnvf82KzQlfxgAl2d1zYbr1u9pr3E3
bandit6@bandit:~$ █
```

Password: *Bmnnvf82KzQlfxgAl2d1zYbr1u9pr3E3*

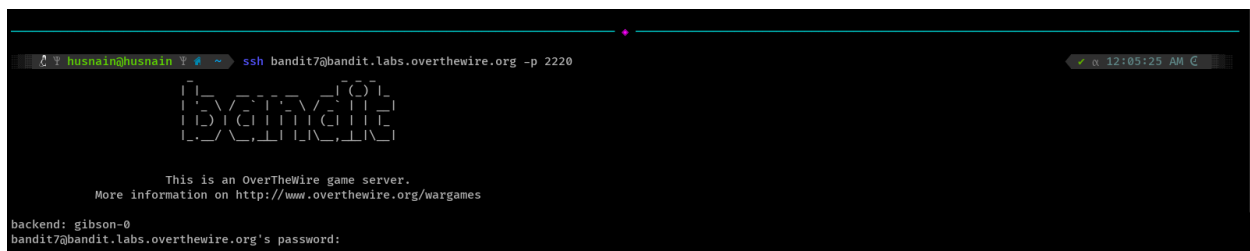
➤ Level 7 → Level 8

Level Goal

The password for the next level is stored in the file data.txt next to the word millionth

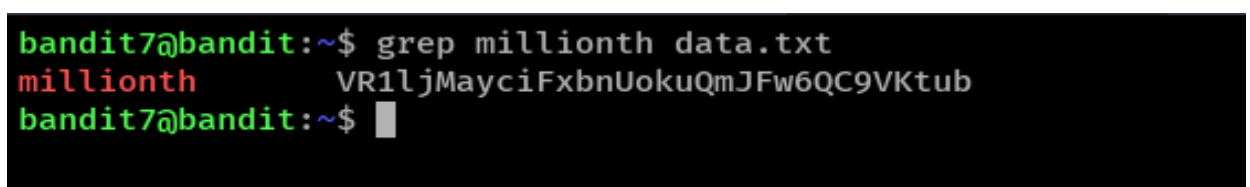
Command Used : ssh, ls, cat, grep

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command "**ssh bandit7@bandit.labs.overthewire.org -p 2220**" to logged in bandit7
- ✓ Used password : Bmnnvf82KzQlfxgAl2d1zYbr1u9pr3E3



```
husnain@husnain:~$ ssh bandit7@bandit.labs.overthewire.org -p 2220
bandit7@bandit:~$
```

- ✓ List the files and folders using "ls"
- ✓ Got the "data.txt" file.
- ✓ Used "cat" command to see the files data but there was too many text in this file.
- ✓ Pressed Ctrl + c to stop but did not worked it show all data in the files.
- ✓ I used "grep" command that searches for lines containing specific text. Since the challenge says the password is next to the word "millionth", grep filters only that matching line.
- ✓ I used command "grep millionth data.txt" and it show the password of the next level.



```
bandit7@bandit:~$ grep millionth data.txt
millionth VR1ljMayciFxbnUokuQmJFw6QC9VKtub
bandit7@bandit:~$
```

Password: VR1ljMayciFxbnUokuQmJFw6QC9VKtub

➤ **Level 8 → Level 9**

Level Goal

The password for the next level is stored in the file data.txt and is the only line of text that occurs only once

Command Used : ssh, ls, sort, uniq with -u flag

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command “**ssh bandit8@bandit.labs.overthewire.org -p 2220**” to logged in bandit8
- ✓ Used password : VR1ljMayciFxbnUokuQmJFw6QC9VKtub

```
➤ husnain@husnain ~ ➡ ssh bandit8@bandit.labs.overthewire.org -p 2220
```

α 12:19:39 AM C

```
[D]X[G]-T-T-V-[O]-L  
[.]X[C][I][K].[E]  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibson-0  
bandit8@bandit.labs.overthewire.org's password:
```

- ✓ List the files and folders using “ls”
- ✓ Got the “data.txt” file same like previous level.
- ✓ Review the level goal that says “password for the next level is stored in the file data.txt and is the only line of text that occurs only once”.
- ✓ Search on google how to check once occur line in a file using cli”, so tells me a command “sort and uniq -u”.
- ✓ Sort command sorts all lines alphabetically (required for uniq to work properly) and Uniq command shows hows only the lines that appear once (-u stands for unique)
- ✓ I used command “sort data.txt | uniq -u” then it show the password of the next level.

```
bandit8@bandit:~$ ls
data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
EjmOSvuAu7sGAHqHVcBDPirRe9T03kx1
bandit8@bandit:~$
```

Password: EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl

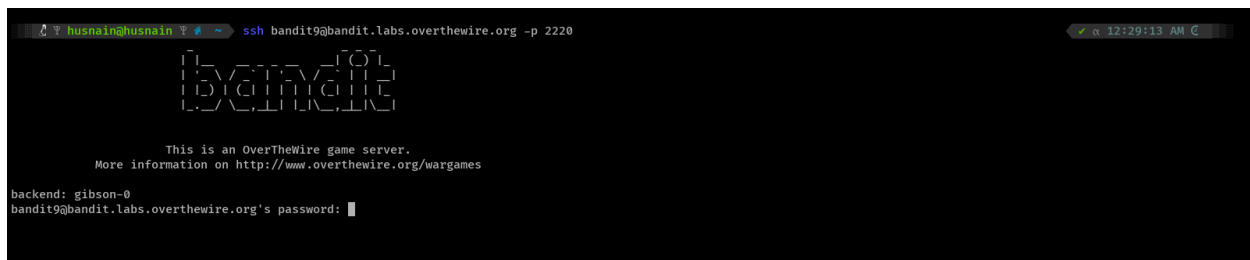
➤ Level 9 → Level 10

Level Goal

The password for the next level is stored in the file data.txt in one of the few human-readable strings, preceded by several '=' characters.

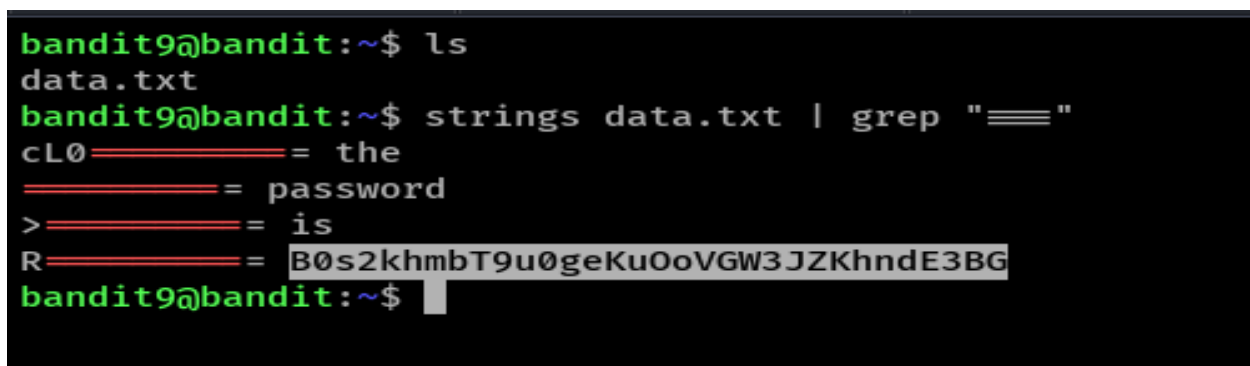
Command Used : ssh, ls, strings, grep

- ✓ I opened a new tab in terminal.
- ✓ Run ssh command "**ssh bandit9@bandit.labs.overthewire.org -p 2220**" to logged in bandit9
- ✓ Used password : EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl



```
➤ husnain@husnain ~$ ssh bandit9@bandit.labs.overthewire.org -p 2220
bandit9@bandit.labs.overthewire.org:~$
```

- ✓ List the files and folders using "ls"
- ✓ Got the "data.txt" file same like previous level.
- ✓ Review the level goal that says "The password for the next level is stored in the file data.txt in one of the few human-readable strings, preceded by several '=' characters".
- ✓ Study about "strings" command that extracts and displays all human-readable text from binary files.
- ✓ Also used "grep" to find the "=" that is mentioned in the level goal.
- ✓ So final command is ""strings data.txt | grep "==" which shows the password of the next level.



```
bandit9@bandit:~$ ls
data.txt
bandit9@bandit:~$ strings data.txt | grep "=="
cL0===== the
===== password
>===== is
R===== B0s2khmbT9u0geKuOoVGW3JZKhndE3BG
bandit9@bandit:~$
```

Password: B0s2khmbT9u0geKuOoVGW3JZKhndE3BG

